

Machine Learning Analysis

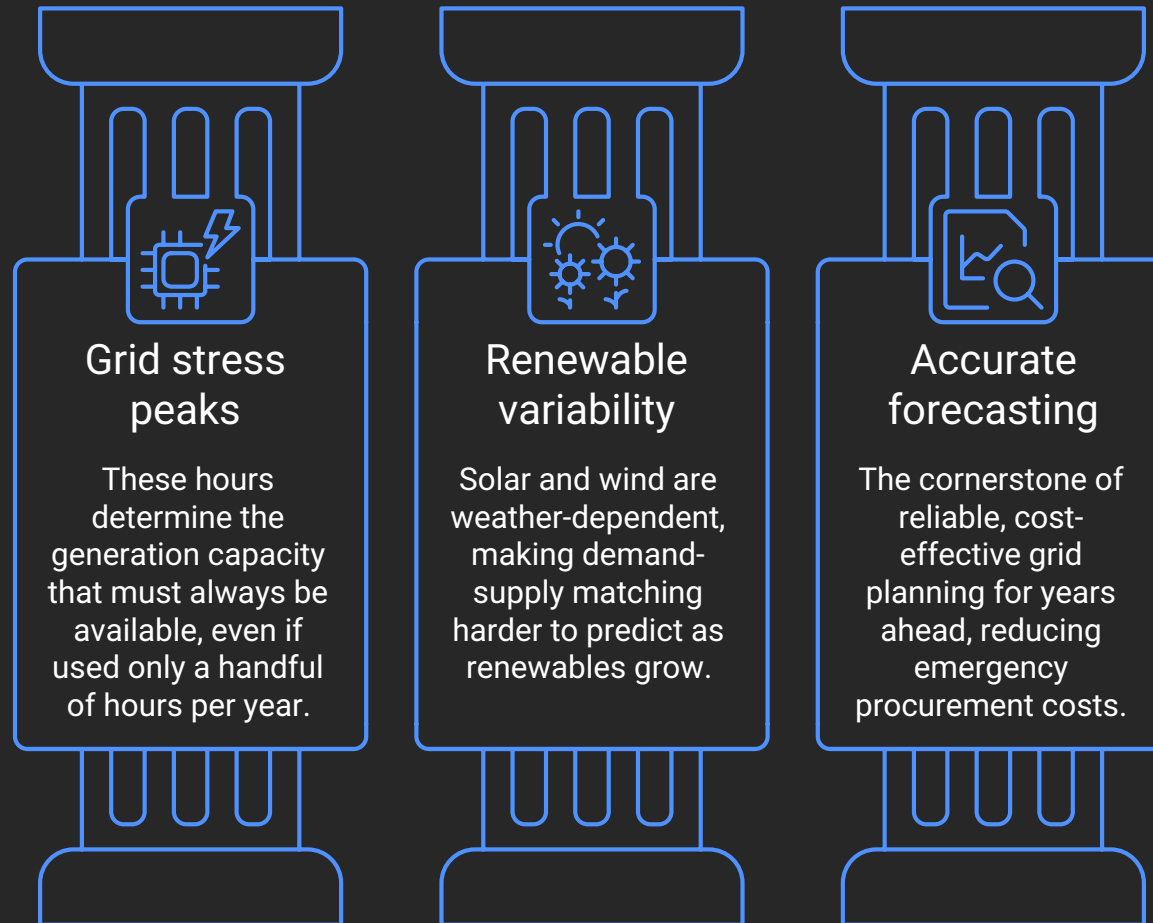
Predicting Peak Electricity Demand in Massachusetts

Grid Reliability and Capacity
Planning through Advanced
Forecasting

By Cathy Kam · September 7,
2026

Why Peak Demand Matters

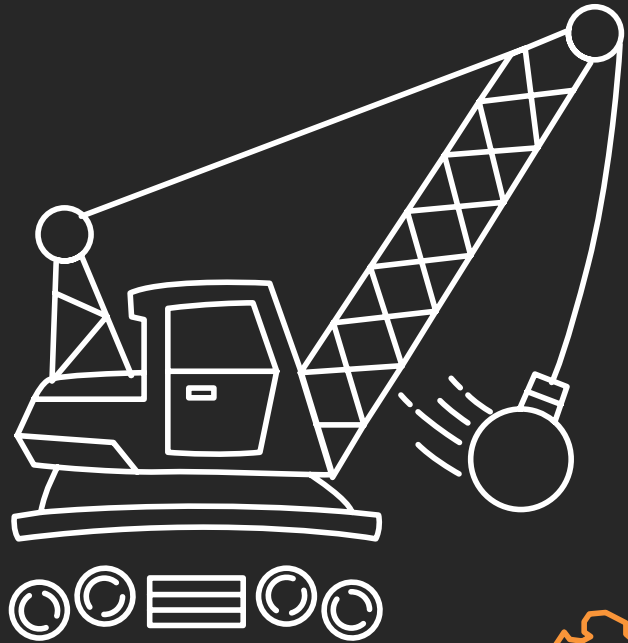
Strategic Context



The Reliability Challenge

Real-Time Balancing

Matching supply and demand



ISO New England must match supply and demand every single hour to maintain grid stability.

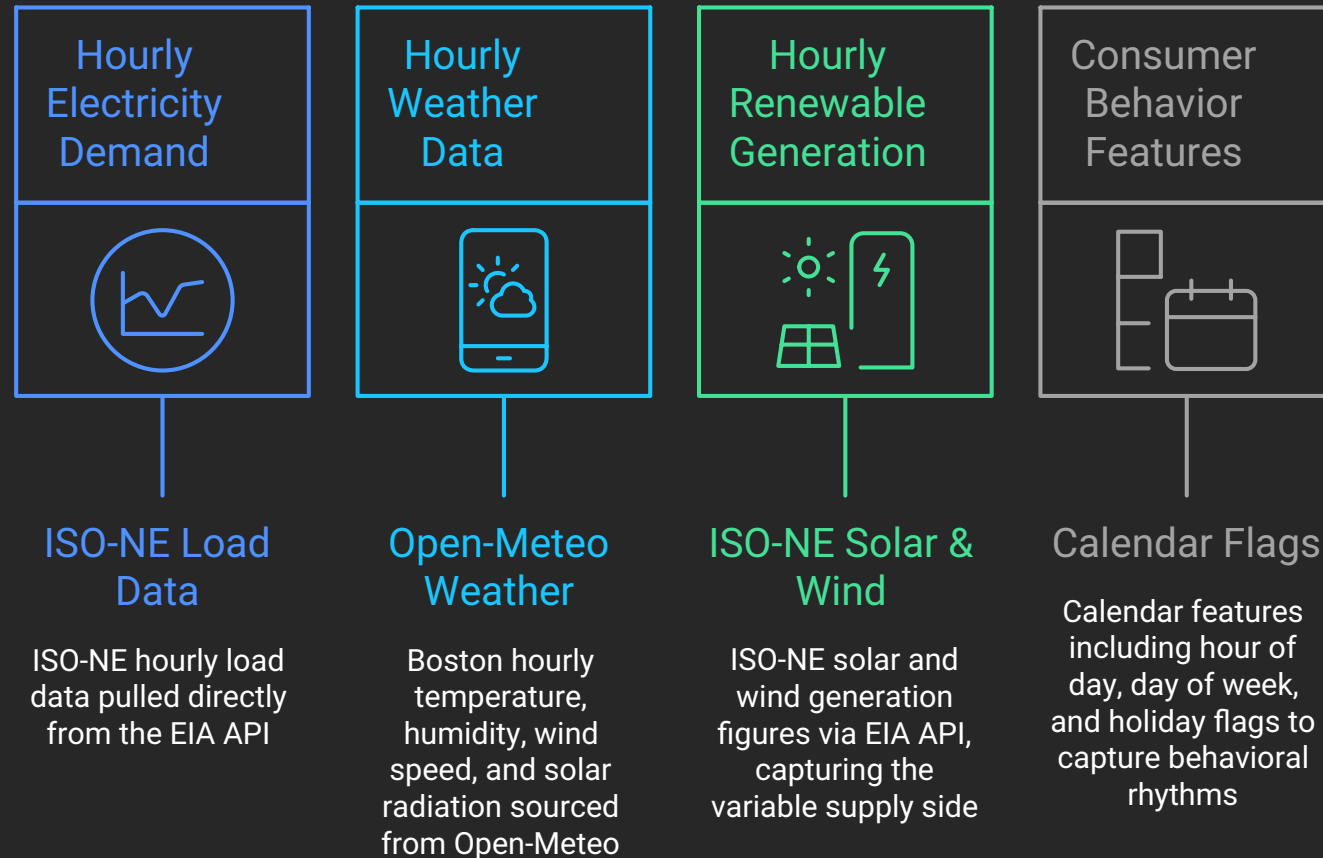
Over-Forecasting

Wasted ratepayer dollars

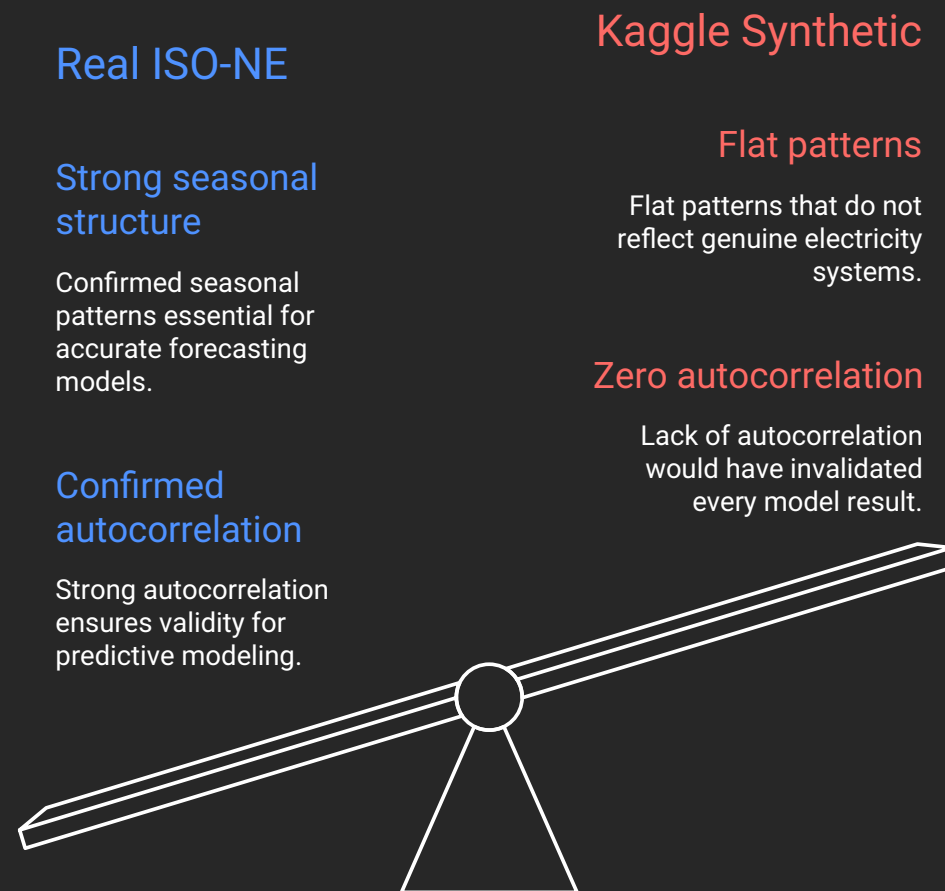
Under-Forecasting

Blackouts and high spot prices

Data Sources for Electricity Demand Prediction



Data Validation Strategy Comparison



Data Validation Strategy Comparison

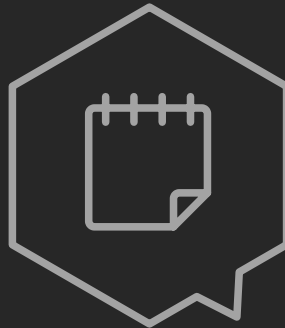
Real ISO-NE — Strong seasonal structure and autocorrelation confirmed, essential for forecasting models.

Feature Engineering Strategy

Transforming raw data into predictive power

Calendar/Behavior

Hour of day, day of week, and weekend/holiday flags for behavioral rhythms.

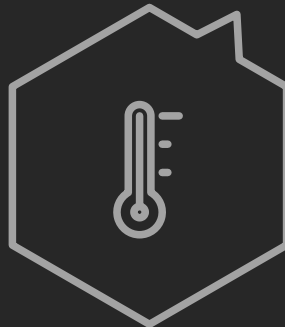


Lag Features

1-hour, 24-hour, and 168-hour (weekly) lags to capture temporal dependencies.

Weather-Derived

Heating degree days, cooling degree days, and temperature interactions.



Rolling Stats

Rolling averages and standard deviations to identify recent trends and volatility.

Model Comparison Results

1.23%

Ensemble MAPE

The best overall performer, combining XGBoost and LightGBM predictions.

151 MW

XGBoost MAE

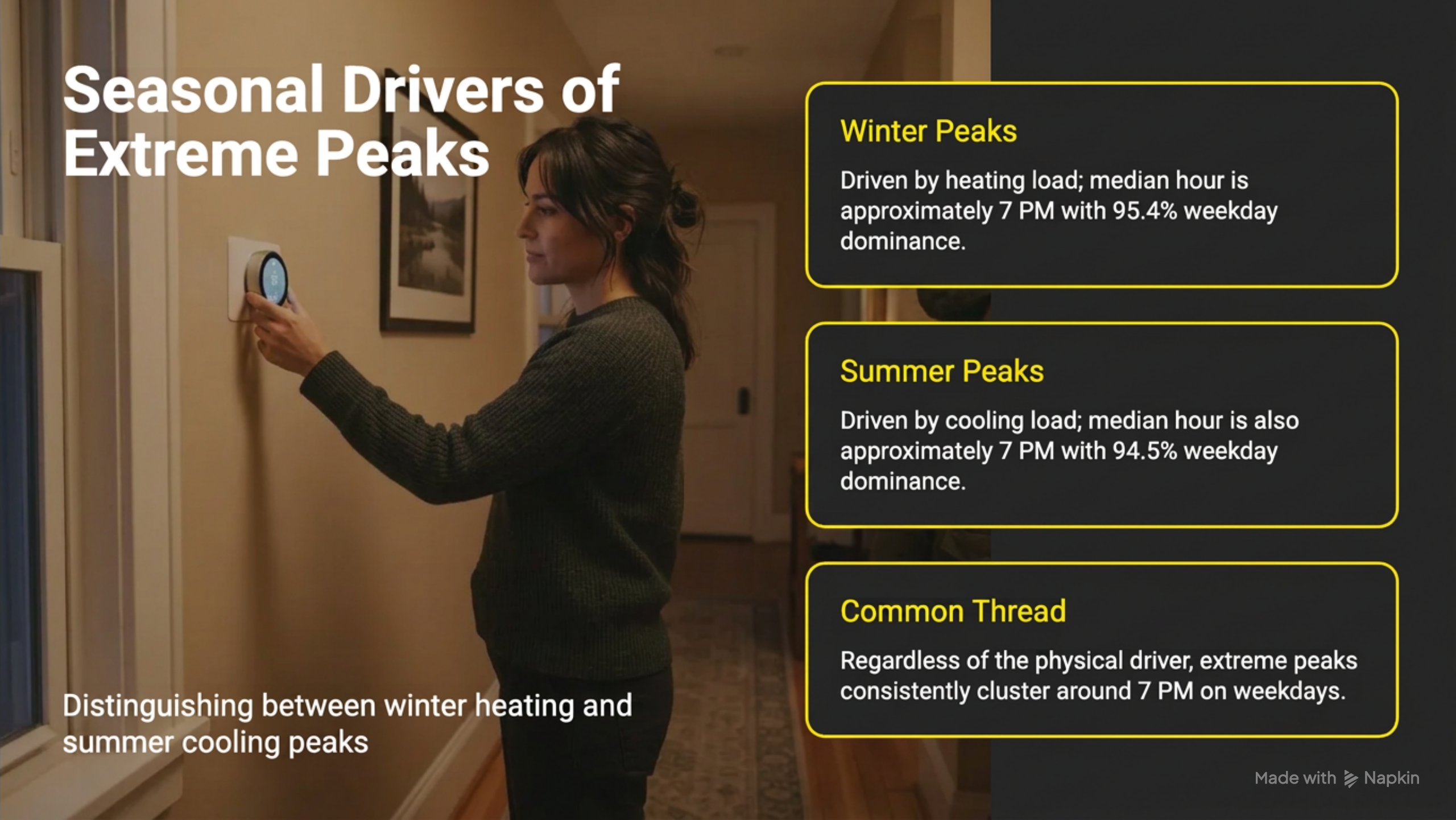
Operationally precise for capacity planning decisions at the Massachusetts scale.

10.47%

Baseline MAPE

The seasonal naive benchmark, which the ensemble model significantly outperforms.

Seasonal Drivers of Extreme Peaks

A woman with dark hair tied back, wearing a green sweater, is standing in a hallway and adjusting a smart thermostat on a wall. The thermostat has a circular display showing the number 70. The background shows a hallway with a framed picture on the wall and a doorway at the end.

Distinguishing between winter heating and summer cooling peaks

Winter Peaks

Driven by heating load; median hour is approximately 7 PM with 95.4% weekday dominance.

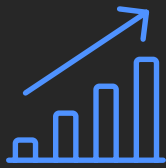
Summer Peaks

Driven by cooling load; median hour is also approximately 7 PM with 94.5% weekday dominance.

Common Thread

Regardless of the physical driver, extreme peaks consistently cluster around 7 PM on weekdays.

Key Takeaways



High-accuracy forecasting

Ensemble MAPE of 1.23% provides a credible basis for capacity commitment.



Structural peaks

Activity timing at 7 PM weekdays amplifies weather stress across both winter and summer.



Data integrity

Real public APIs enabled credible insights that synthetic data would have missed.